

Md. Shaon Khan

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SUMMARY

3rd-year Information Technology undergraduate at the Institute of Information Technology (IIT), Jahangirnagar University with 2+ year of hands-on experience in AI/ML, IoT, and full-stack software development. Developed machine learning applications for healthcare and agriculture, integrated IoT devices with cloud-based software systems, and built end-to-end web applications using modern backend technologies. Seeking an AI/ML or Software Engineering internship to contribute technical skills in real-world product development.

EDUCATION

Institute of Information Technology (IIT), Jahangirnagar University, Savar, Dhaka

B.Sc. in Information Technology | 2022 – Present (3rd Year, 1st Semester)

- CGPA: 3.82 / 4.00 (Up to 4th Semester)

Jahangirnagar University School and College, Savar, Dhaka

Higher Secondary Certificate (HSC) – Science | 2022

- GPA: 5.00 / 5.00

Jahangirnagar University School and College, Savar, Dhaka

Secondary School Certificate (SSC) – Science | 2020

- GPA: 5.00 / 5.00

TECHNICAL SKILLS

- **Languages:** Python, C++, Java, JavaScript, HTML5, CSS3
- **Machine Learning:** TensorFlow, Keras, Scikit-learn, NumPy, Pandas, SciPy, Matplotlib
- **Backend:** FastAPI, Node.js
- **Frontend:** HTML5, CSS3, Bootstrap
- **Database:** MySQL, PostgreSQL
- **IoT:** Arduino Uno, ESP8266, AD8232 ECG Sensor
- **Signal Processing:** Bandpass Filtering, Notch Filtering, Wavelet Denoising, Baseline Wander Removal
- **Deployment:** Render, Netlify, Vercel
- **Tools:** Git, GitHub, VS Code, Jupyter Notebook
- **Operating System:** Windows, Linux

MACHINE LEARNING & SOFTWARE PROJECTS

Machine Learning Projects

HealthBridge — IoT Health Monitoring & Disease Prediction

GitHub: github.com/Md-Shaon-Khan/HealthBridge-with-ECG-Check-Up

- Built a full-stack IoT healthcare platform integrating ECG acquisition with deep learning-based cardiac disease prediction.
- Trained ResNet1D, InceptionTime, and DenseNet121 models on the MIT-BIH, PTB-XL, and CPSC 2018 datasets for eight cardiac condition classes.
- Implemented a multi-stage ECG signal preprocessing pipeline including DC offset removal, baseline wander removal, notch filtering, bandpass filtering, and wavelet denoising.
- Developed a patient-doctor dashboard and integrated a medical chatbot; deployed the FastAPI backend on Render and the frontend on Netlify.

Farmer's Friend (কৃষক বন্ধু) — Paddy Disease Classifier

GitHub: github.com/Md-Shaon-Khan/Farmer-s-Friend

- Developed a deep learning ensemble using EfficientNetB0, ResNet50, and DenseNet121 trained on approximately 49,000 images collected from six datasets.
- Implemented a binary gatekeeper model to reject non-paddy image inputs before disease classification.
- Achieved approximately 97% validation accuracy across five rice leaf disease classes.
- Served as the foundation for ArogyoAI, a Top 100 Finalist project at Infinity AI BuildFest 2026.

Software Engineering Projects

IIT Academic Website — Batch-wise Academic Portal

GitHub: github.com/Md-Shaon-Khan/IIT-Academic-Website-with-Backend

- Developed a batch-wise academic portal for the Institute of Information Technology using FastAPI and MySQL.
- Enabled administrators to manage technical projects, GitHub repositories, supervisors, and image galleries.
- Implemented search functionality for academic projects and designed a responsive interface for examination papers, project archives, and academic calendars.

UniFix — University Complaint Management System

GitHub: github.com/Md-Shaon-Khan/UniFix

- Developed a full-stack complaint management platform with real-time status updates using Socket.io.
- Implemented role-based authentication, issue categorization, and secure database management using MySQL.
- Built dashboards for students and administrators to streamline campus issue reporting and resolution.

TECHNICAL EXPERIENCE

AI/ML Development (1+ Year)

- Built and evaluated machine learning and deep learning models using Scikit-learn, TensorFlow, and Keras across multiple real-world datasets.
- Applied end-to-end ML workflows including EDA, data preprocessing, feature engineering, model evaluation, and hyperparameter tuning.
- Improved model performance through architecture experimentation, parameter optimization, and data preprocessing techniques.

IoT & Signal Processing

- Integrated Arduino Uno, ESP8266, and AD8232 ECG sensors with AI-powered backend systems.
- Implemented ECG signal preprocessing techniques including bandpass filtering, notch filtering, baseline wander removal, and wavelet denoising.

Software Development

- Developed full-stack web applications using FastAPI, Node.js, MySQL, and PostgreSQL.
- Designed RESTful APIs, authentication systems, role-based access control, and real-time features using Socket.io.
- Deployed applications on cloud platforms and managed projects using Git and GitHub.

Hackathon Experience

- **Top 100 Finalist**, Infinity AI BuildFest 2026, BRAC University.
- Contributed as an AI & IoT Developer in developing ArogyoAI, an AI-assisted healthcare solution for predictive diagnostics and decision support.

Competitive Programming

- Solved 400+ algorithmic problems across Codeforces (300+) and LeetCode (100+) using C++ and Python.
- Achieved a peak Codeforces rating of 1182 while strengthening problem-solving, data structures, and algorithm design skills.

Professional Training

Phitron – AI/ML Engineering

- Completed project-based training covering Python programming, statistics, exploratory data analysis (EDA), supervised learning, unsupervised learning, deep learning, CNNs, and modern NLP concepts.
- Implemented hands-on projects covering regression, classification, clustering, computer vision, and end-to-end machine learning workflows using real-world datasets.

CERTIFICATIONS

- Kaggle Learn: Intro to Machine Learning
- Kaggle Learn: Intermediate Machine Learning
- SimpliLearn: PostgreSQL Professional Certification
- HackerRank: Python (Basic)